FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 3

B.Sc. MATHEMATICS HONOURS

MAT5EJ303 (2): CONVEX OPTIMIZATION

Module III: Convex Optimization Problems: Linear, Quadratic and Geometric Programming, Generalized Inequality Constraints, Vector Optimization

Maximum Time: 2 Hours		**Maximum Marks: 70**
	Section A	
	All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)	
1	Define a linear program (LP). Write an LP in standard form and in inequality form.	**3**
2	Express the piecewise-linear minimization problem $\textbf{minimize } \max_{i=1,\ldots,m}\left(a_i^T x + b_i\right)$ as a linear program.	**3**
3	Define a quadratic program (QP) and a quadratically constrained quadratic program (QCQP). What condition on the matrices is needed for them to be convex problems?	**3**
4	Show that the least-squares problem $\textbf{minimize } \|Ax-b\|_2^2$ is an unconstrained QP. Write the objective in the form $(1/2)x^TPx + q^Tx + r$ and give the equations satisfied by the minimizer.	**3**
5	Define a monomial and a posynomial function. State whether each of the following is a monomial, a posynomial or neither: (i) $3x_1^{1/2}x_2^{-1}$, (ii) $x_1 + x_2^2/x_3$, (iii) $x_1 - x_2$.	**3**
6	Write a geometric program (GP) in standard form. With the change of variables $y_i = \log x_i$, write down the equivalent convex form of the GP.	**3**
7	Define a cone program in conic form, with a generalized inequality constraint. Show that a linear program is a conic form problem. What is a linear matrix inequality (LMI)?	**3**
8	Define a second-order cone program (SOCP). Show that the second-order cone constraint $\|Ax+b\|_2 \le c^Tx + d$ is a generalized inequality with respect to the second-order cone.	**3**
9	Define a vector optimization problem $\textbf{minimize (w.r.t. } K\textbf{) } f_0(x)$. Distinguish between an optimal point and a Pareto optimal point.	**3**
10	What is a bi-criterion optimization problem? Define the optimal trade-off curve, and write the regularized least-squares problem as an example.	**3**
	Section B	
	All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)	
11	Give an explicit solution of each of the following LPs. (a) Minimizing a linear function over an affine set: minimize c^Tx subject to $Ax = b$. (b) Minimizing a linear function over a halfspace: minimize c^Tx subject to $a^Tx \le b$, where $a \neq 0$. (c) Minimizing a linear function over a rectangle: minimize c^Tx subject to $l \preceq x \preceq u$, where $l \preceq u$.	**6**
12	Formulate the following problems as LPs. Explain in detail the relation between the optimal solution of each problem and the solution of its equivalent LP. (a) Minimize $\|Ax-b\|_\infty$ (ℓ_∞-norm approximation). (b) Minimize $\|Ax-b\|_1$ (ℓ_1-norm approximation). Here $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ are given.	**6**
13	Show that the linear-fractional program $\textbf{minimize } \dfrac{c^Tx+d}{e^Tx+f}$ subject to $Gx \preceq h$, $Ax = b$, $e^Tx + f > 0$ can be transformed into the LP $\textbf{minimize } c^Ty + dz$ subject to $Gy \preceq hz$, $Ay = bz$, $e^Ty + fz = 1$, $z \ge 0$, with variables y and z.	**6**
14	Verify that $x \in \mathbb{R}^n$ and $y, z \in \mathbb{R}$ satisfy $x^Tx \le yz$, $y \ge 0$, $z \ge 0$ if and only if $\left\|\begin{pmatrix} 2x \\ y-z \end{pmatrix}\right\|_2 \le y+z$, $y \ge 0$, $z \ge 0$. Why is this useful in second-order cone programming?	**6**
15	Express the following problems as convex optimization problems. (a) Minimize $\max\{p(x), q(x)\}$, where p and q are posynomials. (b) Minimize $\exp(p(x)) + \exp(q(x))$, where p and q are posynomials.	**6**
16	Suppose the function $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at a point $x_0 \succ 0$, with $f(x_0) > 0$. How would you find a monomial function $\hat{f} : \mathbb{R}^n \to \mathbb{R}$ such that $f(x_0) = \hat{f}(x_0)$ and, for x near x_0, $\hat{f}(x)$ is very near $f(x)$?	**6**
17	Consider the SDP $\textbf{minimize } c^Tx$ subject to $x_1F_1 + x_2F_2 + \cdots + x_nF_n + G \preceq 0$, with $F_i, G \in \mathbb{S}^k$ and $c \in \mathbb{R}^n$. (a) Suppose $R \in \mathbb{R}^{k\times k}$ is nonsingular. Show that the SDP is equivalent to the SDP with $\tilde{F}_i = R^TF_iR$ and $\tilde{G} = R^TGR$. (b) Suppose there exists a nonsingular R such that $\tilde{F}_i$ and $\tilde{G}$ are diagonal. Show that the SDP is equivalent to an LP.	**6**

18	Show that if x minimizes the scalarized problem $\lambda^T f_0(x)$ over the feasible set, where $\lambda \succ_{K^*} 0$, then x is a Pareto optimal point of the vector optimization problem $\textbf{minimize (w.r.t. } K\textbf{)}\ f_0(x)$. Illustrate with the bi-criterion problem $\textbf{minimize (w.r.t. } \mathbb{R}^2_+\textbf{)}\ \left(\|Ax-b\|_2^2,\ \|x\|_2^2\right)$.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	(a) Show that the constrained least-squares problem $\textbf{minimize}\ \|Ax-b\|_2^2$ subject to $l_i \le x_i \le u_i$ is a QP. Give P, q, r and the inequality constraint matrices. (b) Give an explicit solution of the QCQP $\textbf{minimize}\ c^T x$ subject to $(x-x_c)^T A(x-x_c) \le 1$, where $A \in \mathbb{S}^n_{++}$ and $c \neq 0$.	**10**
20	(a) Show that a geometric program in standard form becomes a convex optimization problem under the change of variables $y_i = \log x_i$, and explain why the constraint functions of the transformed problem are convex. (b) Show that the second-order cone constraint $\|Ax+b\|_2 \le c^T x + d$ is equivalent to the linear matrix inequality $\begin{pmatrix}(c^T x+d)I & Ax+b \\ (Ax+b)^T & c^T x+d\end{pmatrix} \succeq 0$, and express the LP $\textbf{minimize}\ c^T x$ subject to $Gx \preceq h$ as an SDP.	**10**